

Beyond the Bell Curve

From Statistical Fit to Investment Decisions

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Erdos Institute Quant Finance Boot Camp

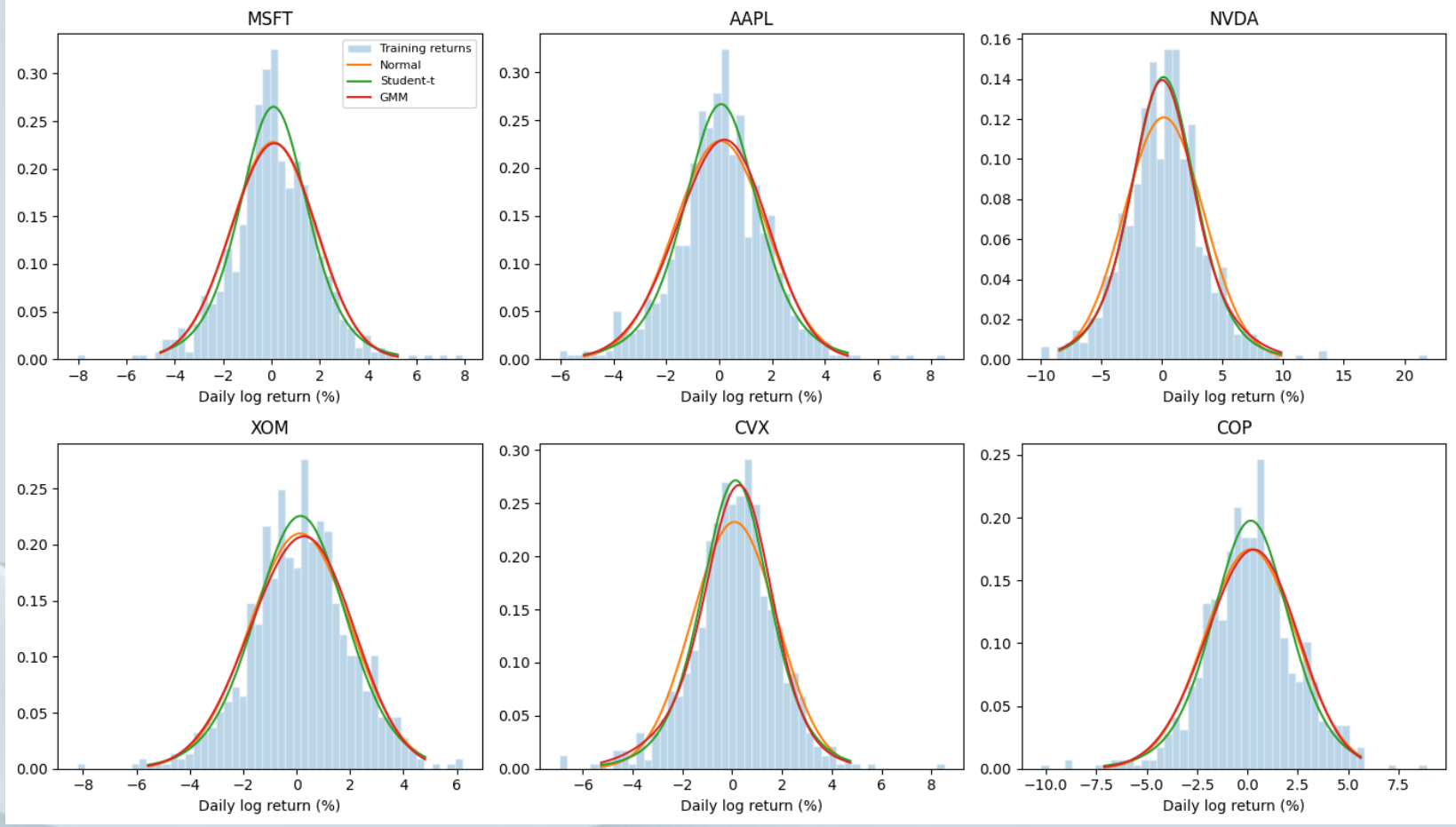
Research Questions

- Which distribution best describes historical stock returns?
- Does a better historical fit lead to more accurate risk forecasts?
- Can risk-based portfolio management improve investment performance, or does it come at an opportunity cost?

Data and research design

- 20 stocks: 10 technology-related and 10 energy
- Training: 2021–2023 | Out-of-sample: 2024–2025 (502 days)
- Fit Normal, Student-t, and two-component Gaussian mixture (GMM)
- Evaluate Bayesian Information Criterion (BIC) , portfolio Value at Risk (VaR), Expected Shortfall (ES), and position-sizing performance

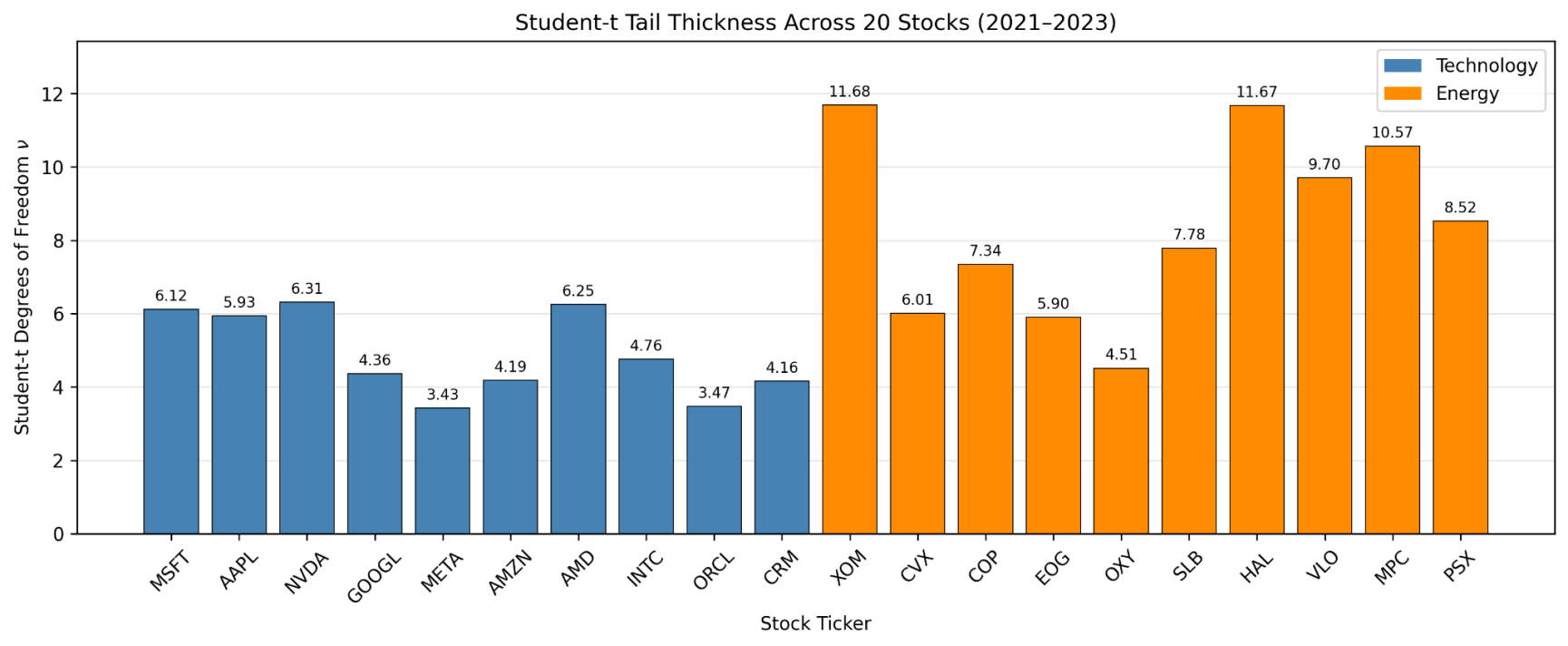
Historical fit: Student-t wins all 20 stocks



Student-t has the lowest Bayesian Information Criterion (BIC) for all 20 stocks.

Student-t model fits the best for all 20 stocks

Tail thickness differs across stocks



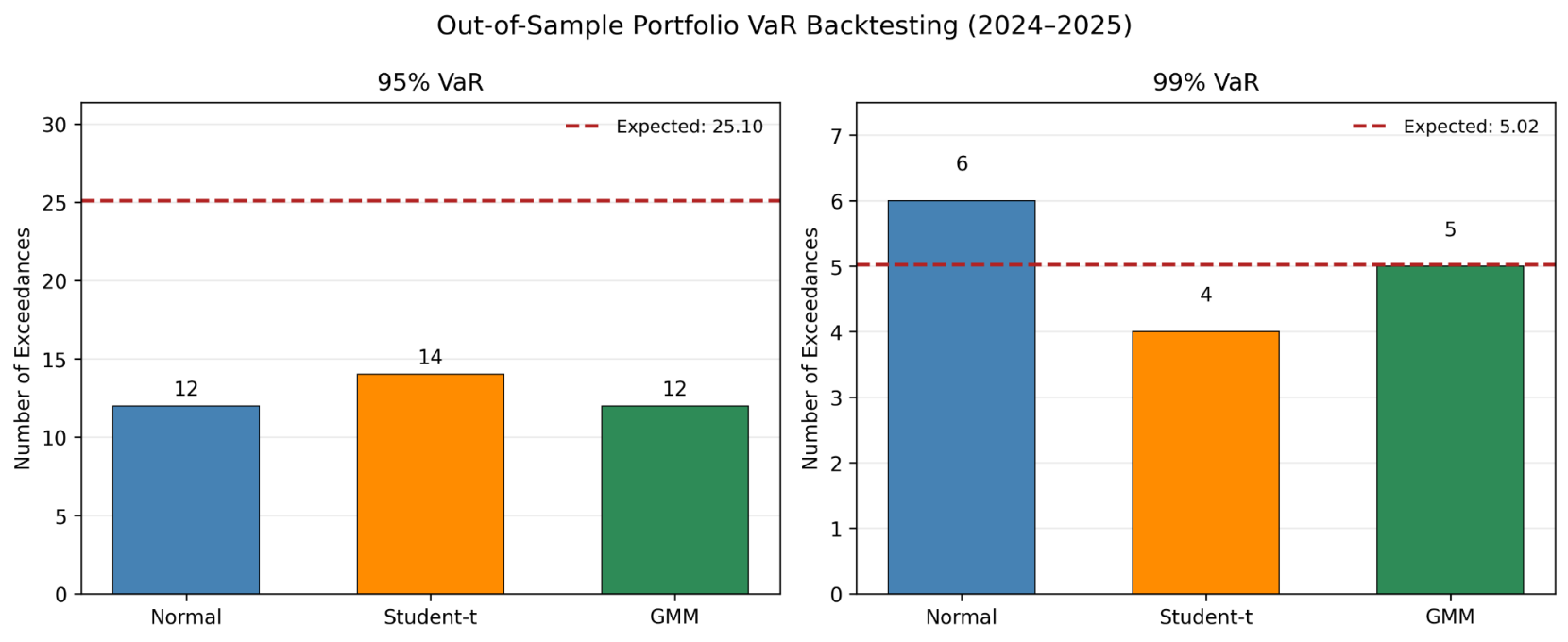
Mean fitted degree-of-freedom parameter ν for student-t model:

Technology 4.90

Energy 8.37

Lower ν means heavier fitted tails.

Out-of-sample VaR: no consistent winner



99% VaR exceedances:

Normal: 6

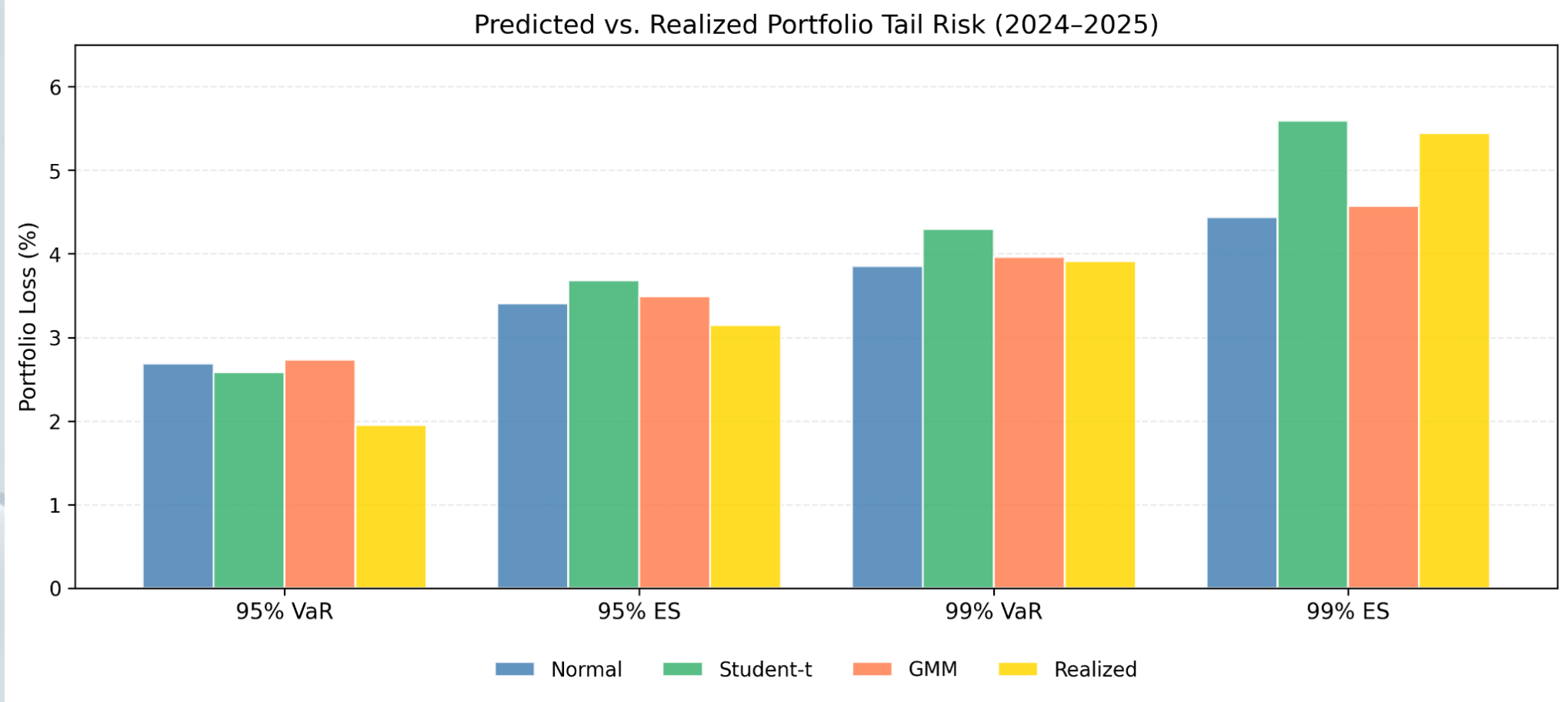
Student-t: 4

GMM: 5

Expected: 5.02

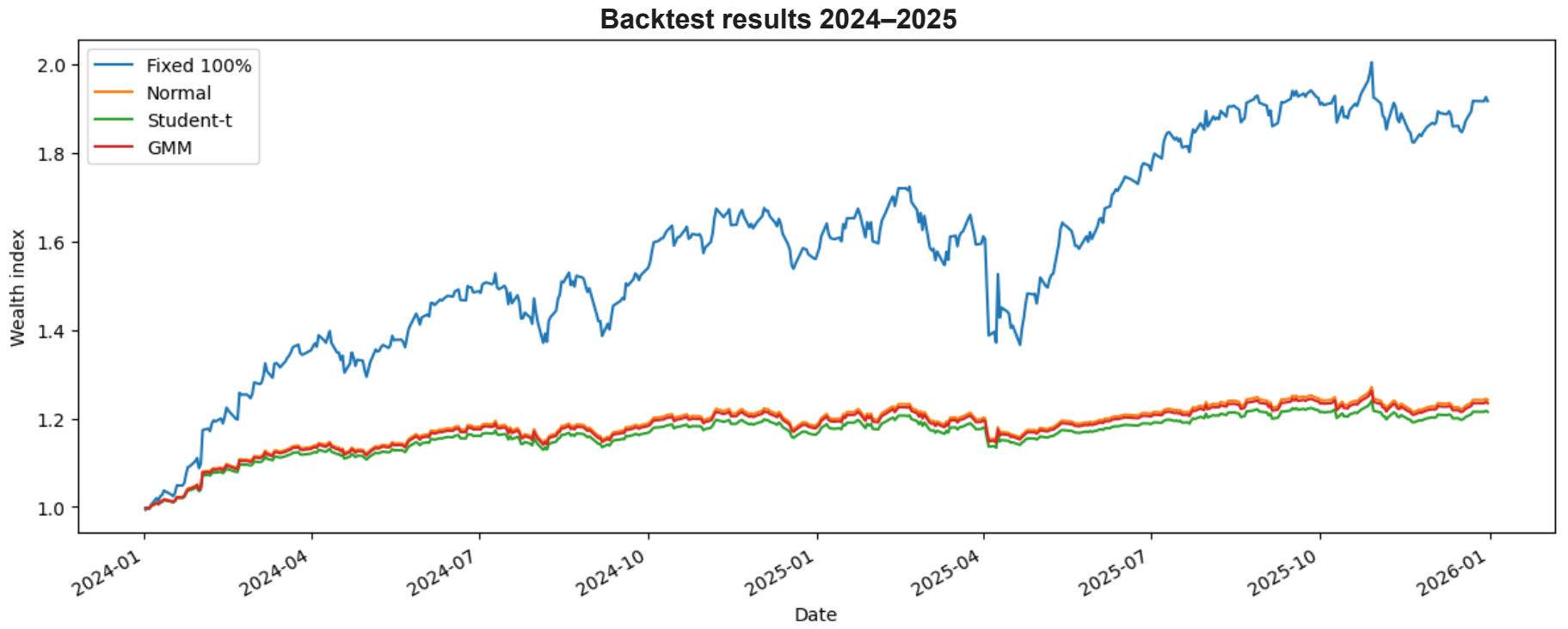
Few extreme days limit strong rankings.

VaR and expected shortfall comparison



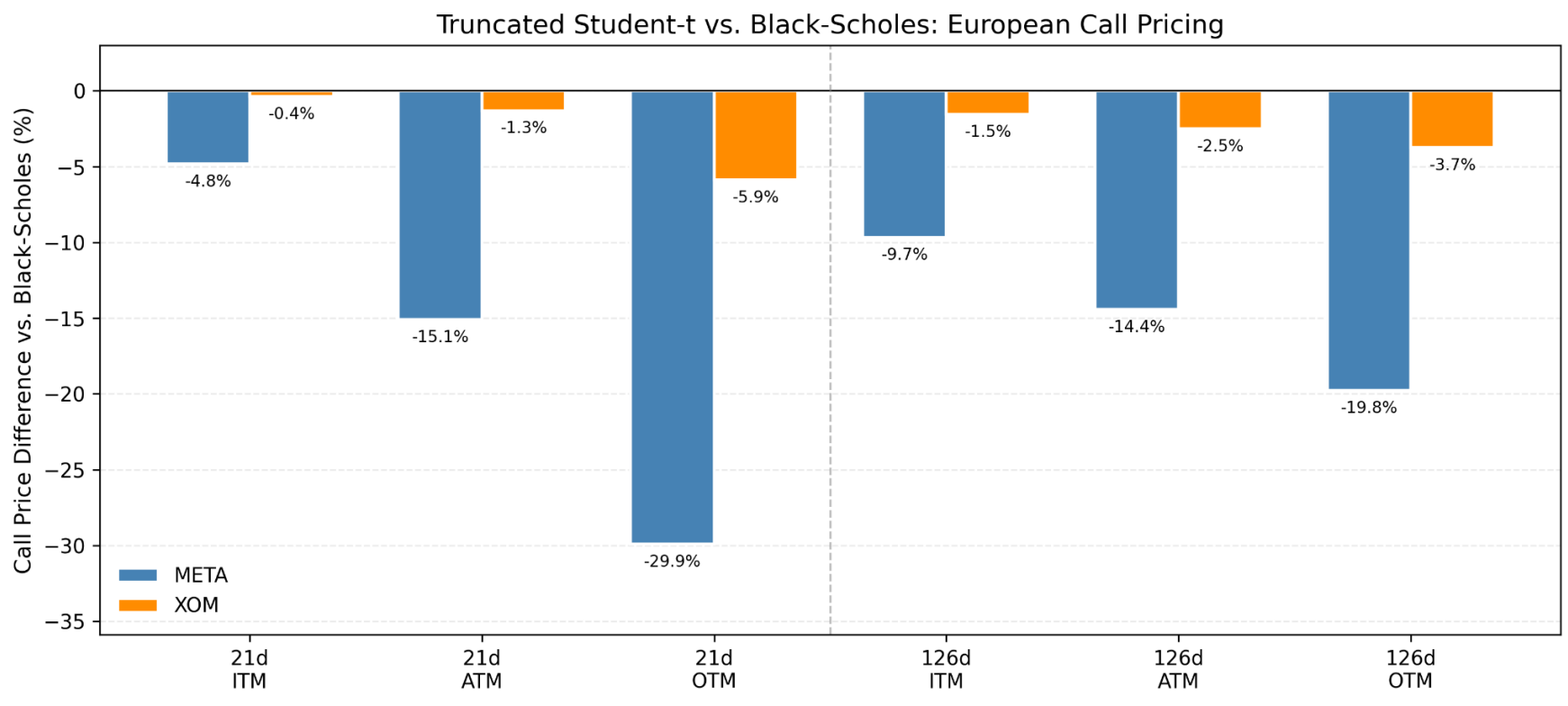
Different models are closest on different risk measures.

Investment test: risk reduction vs. growth



Strategy	Return	Max drawdown	Sharpe	Avg. exposure
Fully invested	91.60%	−20.72%	1.548	100%
Normal VaR	24.17%	−6.78%	1.399	36.98%
Student-t VaR	21.49%	−6.09%	1.399	33.15%
GMM VaR	23.45%	−6.60%	1.399	35.96%

Theoretical European call pricing



Truncated Student-t prices differ more from Black–Scholes for META than for XOM.

No market option quotes were used.

Conclusion: protection has an opportunity cost

- Student-t fits historical return best, but does not consistently improve out-of-sample risk forecasts.
- VaR-based strategies reduce drawdowns, but sacrifice 67–70% of cumulative return.
- All three risk-controlled strategies have lower Sharpe ratios (~1.40) than the benchmark (1.55).
- Better statistical models do not guarantee better investment decisions. Risk protection comes at an opportunity cost.

No risk, no gain !